Teacher:	Ray Joshua Balingkit	**Subject:**	Computing
Week:	5 (Term 1)	**Year:**	7
Link:		**Cover:**	

LESSON OVERVIEW

What should students know/do by the end of this week's lessons?

Students will learn to encode and decode a short message accurately using an agreed binary character code. They will also learn to compare how text, bitmap images and sampled sound use numeric codes to represent different content. Through teacher demonstrations, discussion and practical activities, students will practise these skills and explain their ideas.

End-of-Week Assessment/Check: How will you know they got it?

Use teacher observation, pupil explanations and the completed practical or paper activity. Assess the accurate round trip, clear byte grouping and an explanation of one correction. Check a correct representation for each type and a comparison that states what the codes mean. Note pupils who need further guided practice.

LESSON SLOTS - MONDAY AND TUESDAY

Lesson	Date	Learning Objective	The Lesson Flow	Differentiation
1	21 September 2026	By the end of this lesson, students will be able to encode and decode a short message accurately using an agreed binary character code.	**H:** Show two similar encoded words and ask how one changed bit could affect decoding. **ID:** Review the supplied character table, fixed eight-bit groups and the difference between characters and number values. **WD:** Encode 'cat', group each character correctly and decode the result to check it against the original. **YD:** Pupils exchange short non-personal messages, decode them and correct a deliberate grouping or copying error. **C:** Assess the accurate round trip, clear byte grouping and an explanation of one correction.	**Support:** Provide place-value grids, carry rows, a character-code table and labelled diagrams. Model one step at a time and allow spoken reasoning before writing. **Extension:** Ask pupils to justify a method, identify a deliberate error or test an eight-bit boundary, then verify the result using another representation.
2	22 September 2026	By the end of this lesson, students will be able to compare how text,	**H:** Show a character-code table, a pixel grid and a set of sound samples.	**Support:** Provide place-value grids, carry rows, a character-code table and labelled

		bitmap images and sampled sound use numeric codes to represent different content.	**ID:** Review the common use of binary and the different meanings of the stored numbers. **WD:** Build a comparison table linking text to characters, images to pixel colours and sound to measurements over time. **YD:** Pupils complete the table, decode a small example and explain why more uncompressed samples or pixels usually need more data. **C:** Check a correct representation for each type and a comparison that states what the codes mean.	diagrams. Model one step at a time and allow spoken reasoning before writing. **Extension:** Ask pupils to justify a method, identify a deliberate error or test an eight-bit boundary, then verify the result using another representation.
3				Support: Extension:
4				Support: Extension:

MATERIALS NEEDED FOR THE WEEK

- Oxford International Lower Secondary Computing Student Book 7 printed pages 22-29
- Teacher computer and display
- Classroom computers or laptops and the class save folder
- Binary place-value grids, carry rows, character-code tables and pixel or sampling sheets
- Whiteboard and markers
- Student notebooks, pencils and task sheets
- Teacher reference: Unicode Consortium glossary, https://unicode.org/glossary/



END-OF-WEEK REFLECTION

- What worked well?

- What flopped or confused students?

- What needs to be re-taught or adjusted next week?